

PRAVEEN K

SOC Analyst | Junior Threat Intelligence Analyst | Cybersecurity Analyst

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PROFESSIONAL SUMMARY

Entry-level SOC Analyst with proven hands-on depth across the full detection-to-response lifecycle — built a home lab from scratch simulating real-world MITRE ATT&CK kill chain scenarios including brute force, lateral movement, and C2 beaconing. Engineered Pherion, a Python-based IDS with 37+ MITRE-mapped detection rules and Splunk integration, achieving ~30% false positive reduction through ML-based anomaly detection. Skilled in IOC/IOA analysis, PCAP investigation, Sigma rule authoring, alert triage, and SIEM-based incident correlation. Certified across Cisco, Fortinet, and Palo Alto Security Operations. Targeting L1 SOC Analyst roles at MSSPs and GCCs to contribute to 24/7 security operations and incident response.

TECHNICAL SKILLS

SIEM & Monitoring : Splunk (log ingestion, alert rules, dashboards, triage), Security Event Monitoring, Log Analysis, Alert Investigation, False Positive Reduction, Incident Ticketing.

Security Tools : Wireshark, Nmap, Burp Suite, Nessus, Metasploit, Kali Linux

Threat Detection & Response : IDS/IPS, IOC/IOA Analysis, Threat Hunting, Threat Intelligence, Malware Analysis, Incident Response, PCAP Analysis, Vulnerability Assessment, RCA (Root Cause Analysis), Alert Escalation, Playbook/Runbook Execution.

Frameworks & Standards : MITRE ATT&CK, Cyber Kill Chain, OWASP Top 10, NIST Cybersecurity Framework (awareness), IAM Fundamentals, CVSS Scoring (awareness).

Networking & Protocols : TCP/IP, DNS, HTTP/HTTPS, VPN, Firewalls, ARP, DHCP, Packet Analysis, Network Scanning, Port Enumeration.

OS & Platforms : Kali Linux, Windows Security, Linux CLI, VirtualBox, Virtual Lab Environments.

Programming & Scripting : Python (security scripting, automation), Bash/Shell Scripting, PowerShell (basics).

Concepts : AES-256 Encryption, Steganography, Zero Trust Architecture, EDR (CrowdStrike/SentinelOne), SOAR.

PROJECTS

Pherion – AI-Powered IDS & SOC Monitoring Platform

Tech Stack: Python, Scapy, Scikit-learn, Pandas, SQLite, JSON, Splunk

- Engineered 37+ detection rules mapped to MITRE ATT&CK TTPs (T1046 – Port Scan, T1071 – C2 Beaconing, T1110 – Brute Force, T1557 – ARP Spoofing, T1048 – DNS Tunneling, T1021 – Lateral Movement) for structured threat intelligence reporting.
- Integrated ML-based anomaly detection with auto-tuning thresholds, reducing false positive alert noise by ~30% in lab testing – improving analyst triage efficiency.
- Generated SIEM-ready JSON logs ingested into Splunk, enabling real-time alert investigation and SOC workflow simulation with dashboard visibility.

Praxion – USB Malware Scanner & Threat Detection Tool

Tech Stack: Python, YARA, PE File Analysis, VirusTotal API, ssdeep

- Automated IOC identification using VirusTotal API and ssdeep fuzzy hashing, enabling faster malware triage and threat classification decisions.
- Built YARA rule matching engine with heuristic scanning and PE inspection for detecting known and unknown malware families on USB devices.
- Implemented forensic-grade logging and secure quarantine system, producing structured evidence output suitable for incident documentation.

PRISM – SIEM Rule Generator from Raw Logs

Tech Stack: Python, Rich, Regex, Sigma YAML, Splunk SPL, JSON, MITRE ATT&CK

- Architected a regex parsing pipeline for auth.log, Apache access logs, and Windows Security events – detecting failed logins, port scan surges, and privileged execution attempts per source IP against defined thresholds.
- Engineered automated rule translation layer converting raw log insights into deployable Sigma YAML and Splunk SPL detection rules, with a built-in validation module cross-referencing generated rules against sample malicious logs before deployment.
- Designed a MITRE ATT&CK mapping engine assigning tactics and CVE identifiers to extracted artifacts, with normalized JSON output for universal SIEM ingestion.

HOME LAB – SOC ANALYST SIMULATION ENVIRONMENT

Tech Stack: VirtualBox, Kali Linux, Windows 10, Splunk Enterprise, Wireshark

- Built a segmented dual-VM SOC lab (Kali Linux / Windows 10 + Splunk) to simulate real-world attack scenarios including T1110 (Brute Force), T1078 (Valid Accounts), T1046 (Port Scan), and T1021 (Lateral Movement).
- Performed L1 analyst triage: detected alerts → investigated event logs → correlated IOCs → escalated findings → documented incidents in structured ticketing format (simulating ServiceNow/Jira workflow).
- Created custom Splunk detection rules with false positive tuning; followed NIST IR playbook (Identify → Contain → Eradicate → Recover → Document) for each simulated incident.

THREAT INTELLIGENCE & CVE RESEARCH

CVE-2021-44228 | Log4Shell RCE Analysis

- Analyzed Log4Shell RCE flaw in Apache Log4j (v2.0–2.14.1) where malicious JNDI strings in HTTP headers trigger remote code execution; mapped attack chain: Initial Access (T1190) → Code Execution (T1059) → C2 Callback (T1071).
- Built Splunk query to detect JNDI patterns in web logs and alert on unexpected outbound LDAP traffic – simulating SOC L1 triage workflow.
- Documented patch (Log4j 2.15.0), WAF rules, and input sanitization as mitigations in a structured SOC analyst incident brief.

Scattered Spider (UNC3944) – Threat Actor Analysis

- Analyzed the 2023 MGM/Caesars campaign – mapped full attack chain: MFA fatigue (T1621) → credential theft (T1078) → Okta abuse → lateral movement (T1021) → ransomware (T1486) across MITRE ATT&CK, producing a SOC L1 threat intelligence brief.
- Built Splunk detection logic for MFA fatigue patterns and anomalous Okta logins; documented mitigations including adaptive MFA, UEBA alerting, and zero-trust access review aligned to NIST IR playbook.

HANDS-ON TRAINING & INTERNSHIP

Hackpvt Technology pvt Ltd – Cybersecurity Internship (1 Month)

- Performed web application security testing using Burp Suite; identified SQLi, XSS, and IDOR vulnerabilities mapped to OWASP Top 10.
- Analyzed network traffic for suspicious packets/IOCs using Wireshark; conducted network scanning and service enumeration with Nmap to identify open ports and attack surfaces.
- Executed controlled exploitation using Metasploit on vulnerable lab machines; documented findings and remediation steps including privilege escalation analysis.

Vital Skills x Techkriti – IIT Kanpur - Ethical Hacking & Network Security Internship (1 Month)

- Practiced full penetration testing lifecycle (Recon → Scan → Exploit → Post-Exploit) aligned with Cyber Kill Chain; simulated MITM, brute-force, and network sniffing attacks to understand attacker TTPs and defensive controls.
- Performed vulnerability assessments using Nmap and Burp Suite; analyzed PCAP files in Wireshark to trace attack paths and produced risk-rated reports with remediation notes.

EDUCATION

B.E. Computer Science and Engineering

St.Michael College of Engineering and Technology - Kalayarkoil

2022- 2026

CERTIFICATIONS

- Cisco Networking Academy – Ethical Hacker
- Fortinet Training Institute – NSE 3 – Associate in Cybersecurity
- Cisco Networking Academy – Network Defence
- Infosys Springboard – IoT & Electronics
- Palo Alto Networks – Security Operations Fundamentals